

Differential Equations

Quiz Solutions & Comprehensive Explanations

1. The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^2 + y^2 = 8$ are respectively:

- (a) 1, 1
- (b) 2, 3
- (c) 1, 3
- (d) 1, 2

Correct Answer: (d) 1, 2

Explanation: The *order* of a differential equation is the order of the highest derivative present, which is $\frac{dy}{dx}$ (Order = 1). The *degree* is the highest power of the highest-order derivative after removing radicals/fractions involving derivatives. The term $\left(\frac{dy}{dx}\right)^2$ has power 2 (Degree = 2). Hence, (1, 2).

2. The order and degree of $\sec x = y\sqrt{\frac{dy}{dx}}$ are respectively:

- (a) 2, 1
- (b) 1, 2
- (c) 2, 2
- (d) 1, 1

Correct Answer: (b) 1, 2

Explanation: To find the degree, eliminate the radical by squaring both sides: $\sec^2 x = y^2 \frac{dy}{dx}$. The highest derivative present is $\frac{dy}{dx}$, which is of order 1, and its power is 2 in polynomial form after clearing radicals. Thus, order = 1, degree = 2.

3. The order and degree of $\left(1 + \frac{dy}{dx}\right)^{1/2} = c \frac{d^2 y}{dx^2}$ are respectively:

- (a) 2, 2
- (b) 3, 5
- (c) 2, 3
- (d) 1, 3

Correct Answer: (a) 2, 2

Explanation: Square both sides to remove fractional exponents: $1 + \frac{dy}{dx} = c^2 \left(\frac{d^2 y}{dx^2}\right)^2$. The highest derivative is $\frac{d^2 y}{dx^2}$ (order = 2), and its highest power is 2 (degree = 2).

4. The order and degree of $\dot{x} + \log(1+t)\dot{x} = 0$ are respectively:

- (a) 1, 1
- (b) 2, 1
- (c) 2, 2
- (d) 3, 1

Correct Answer: (a) 1, 1

Explanation: Here $\dot{x} = \frac{dx}{dt}$. Factoring gives $\dot{x}(1 + \log(1 + t)) = 0$. The highest derivative is $\dot{x}$ (order 1), raised to exponent 1 (degree 1).

5. Determine the differential equation whose general solution is $y = cx^4$:

- (a) $\frac{dy}{dx} = 4x^3$
- (b) $\frac{dy}{dx} = y \left(\frac{4}{x}\right)$
- (c) $\frac{dy}{dx} = \frac{4y}{x}$
- (d) $x^2 \frac{dy}{dx} = 4y$

Correct Answer: (c) $\frac{dy}{dx} = \frac{4y}{x}$

Explanation: Differentiate $y = cx^4$ with respect to x : $\frac{dy}{dx} = 4cx^3$. Since $c = \frac{y}{x^4}$, substitute c : $\frac{dy}{dx} = 4 \left(\frac{y}{x^4}\right) x^3 = \frac{4y}{x}$.

6. Determine the differential equation whose general solution is $y = ce^{2x}$:

- (a) $\frac{dy}{dx} + 3y = 0$
- (b) $\frac{dy}{dx} - 2y = 0$
- (c) $\frac{dy}{dx} = 3y$
- (d) $\frac{dy}{dx} = \frac{y}{3}$

Correct Answer: (b) $\frac{dy}{dx} - 2y = 0$

Explanation: Differentiating $y = ce^{2x}$ yields $\frac{dy}{dx} = 2ce^{2x} = 2y$. Rearranging gives $\frac{dy}{dx} - 2y = 0$.

7. Determine the differential equation whose general solution is $y = A \cos x + B \sin x$:

- (a) $y'' + y = 0$
- (b) $y' + y = 0$
- (c) $y'' - y = 0$
- (d) $y' - y = 0$

Correct Answer: (a) $y'' + y = 0$

Explanation: $y' = -A \sin x + B \cos x$, and $y'' = -A \cos x - B \sin x = -y$. Thus $y'' + y = 0$.

8. Solve $4y dx + x dy = 0$:

- (a) $yx^3 = c$
- (b) $x^4y = c$

- (c) $yx = c$
 (d) $xy^4 = c$

Correct Answer: (b) $x^4y = c$

Explanation: Separate variables: $\frac{4}{x}dx + \frac{1}{y}dy = 0$. Integrate both sides: $4\ln|x| + \ln|y| = C \implies \ln(x^4y) = C \implies x^4y = c$.

9. Solve $t\frac{dy}{dt} + 5y = 0$, with initial condition $y(1) = 1$:

- (a) $y = t^5$
 (b) $y = \frac{5}{t}$
 (c) $y = \frac{1}{t^5}$
 (d) $y = 5t$

Correct Answer: (c) $y = \frac{1}{t^5}$

Explanation: Separating variables yields $\frac{dy}{y} = -5\frac{dt}{t} \implies \ln y = -5\ln t + C \implies y = Ct^{-5}$. Using $y(1) = 1$ gives $C = 1$, so $y = t^{-5} = \frac{1}{t^5}$.

10. Find the particular solution of $e^x\frac{dy}{dx} = 4$, given $y = 3$ when $x = 0$:

- (a) $y = -4e^{-x} + 7$
 (b) $y = 4e^{-x} + 7$
 (c) $y = 4e^x + 7$
 (d) $y = e^x + 7$

Correct Answer: (a) $y = -4e^{-x} + 7$

Explanation: $\frac{dy}{dx} = 4e^{-x} \implies y = -4e^{-x} + C$. At $x = 0, y = 3$: $3 = -4(1) + C \implies C = 7$. Thus $y = -4e^{-x} + 7$.

11. Solve $x\frac{dy}{dx} - 3y = 4x$:

- (a) $y = Cx^3 - 2x$
 (b) $y = Ce^{3x} - 2x$
 (c) $y = x + C$
 (d) $y = 2x + Cx^3$

Correct Answer: (a) $y = Cx^3 - 2x$

Explanation: Standard linear form: $\frac{dy}{dx} - \frac{3}{x}y = 4$. Integrating factor: $I(x) = e^{\int -3/x dx} = x^{-3}$. Integrating: $yx^{-3} = \int 4x^{-3}dx = -2x^{-2} + C \implies y = -2x + Cx^3$.

12. Which of the following is a First Order Homogeneous ODE?

- (a) $\frac{dy}{dx} = x^2 + y$
 (b) $\frac{dy}{dx} = y + x^2$
 (c) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$

(d) $\frac{dy}{dx} = x + y$

Correct Answer: (c) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$

Explanation: A differential equation $\frac{dy}{dx} = f(x, y)$ is homogeneous of degree 0 if $f(tx, ty) = f(x, y)$. For option (c), $\frac{(tx)^2+(ty)^2}{(tx)(ty)} = \frac{t^2(x^2+y^2)}{t^2xy} = \frac{x^2+y^2}{xy}$, which is homogeneous.

13. Which of the following is NOT a First Order Homogeneous ODE?

(a) $\frac{dy}{dx} = \frac{x}{y}$

(b) $\frac{dy}{dx} = \frac{x+y}{x}$

(c) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$

(d) $\frac{dy}{dx} = \frac{x+y+1}{x}$

Correct Answer: (d) $\frac{dy}{dx} = \frac{x+y+1}{x}$

Explanation: The constant term +1 in the numerator breaks homogeneity because $\frac{tx+ty+1}{tx} \neq \frac{x+y+1}{x}$.

14. Which of the following is a First Order Exact ODE?

(a) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$

(b) $(2x^3y)dx + (y^3 + x)dy = 0$

(c) $(y^3 + x)dx + (x^3 + 4y)dy = 0$

(d) $(2xy^2 + y)dx + (x^3 + y)dy = 0$

Correct Answer: (a) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$

Explanation: For $Mdx + Ndy = 0$, exactness requires $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Here $M = 3x^2y + y^2 \implies \frac{\partial M}{\partial y} = 3x^2 + 2y$. $N = x^3 + 2xy \implies \frac{\partial N}{\partial x} = 3x^2 + 2y$. Since they match, it is exact.

15. Which of the following is NOT a First Order Exact ODE?

(a) $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

(b) $(xy + 1)dx + (\frac{1}{2}x^2 + y)dy = 0$

(c) $(y^3 + x)dx + (x^3 + y)dy = 0$

(d) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$

Correct Answer: (c) $(y^3 + x)dx + (x^3 + y)dy = 0$

Explanation: For (c), $M = y^3 + x \implies \frac{\partial M}{\partial y} = 3y^2$, whereas $N = x^3 + y \implies \frac{\partial N}{\partial x} = 3x^2$. Since $3y^2 \neq 3x^2$, the equation is not exact.

16. Particular solution of $y = A \cos 2x + B \sin 2x$, given $y(\pi/4) = 3$ and $y'(\pi/4) = 8$:

(a) $y = 4 \cos 2x + 3 \sin 2x$

(b) $y = -4 \cos 2x + 3 \sin 2x$

(c) $y = -4 \cos 2x - 3 \sin 2x$

(d) $y = 4 \cos 2x - 3 \sin 2x$

Correct Answer: (b) $y = -4 \cos 2x + 3 \sin 2x$

Explanation: $y(\pi/4) = A \cos(\pi/2) + B \sin(\pi/2) = B = 3$.

$y' = -2A \sin 2x + 2B \cos 2x \implies y'(\pi/4) = -2A(1) + 2B(0) = -2A = 8 \implies A = -4$.

Thus $y = -4 \cos 2x + 3 \sin 2x$.

17. Differential equation whose general solution is $y = Ae^{2x} + Be^{-x}$:

(a) $y'' + y' - 2y = 0$

(b) $y'' - y' + 2y = 0$

(c) $y'' - y' - 2y = 0$

(d) $y'' + y' + 2y = 0$

Correct Answer: (c) $y'' - y' - 2y = 0$

Explanation: Roots of characteristic equation are $m_1 = 2, m_2 = -1$. The characteristic equation is $(m - 2)(m + 1) = 0 \implies m^2 - m - 2 = 0$. This corresponds to $y'' - y' - 2y = 0$.

18. Solve $\frac{dy}{dx} = \frac{x+y}{2x}$:

(a) $(x + y)^2 = Ax^3$

(b) $(1 + y)^2 = Ax^3$

(c) $(y - x)^2 = Ax$

(d) $(1 + y)^2 = Ax^2$

Correct Answer: (c) $(y - x)^2 = Ax$

Explanation: Substitute $y = vx \implies v + x \frac{dv}{dx} = \frac{1+v}{2} \implies x \frac{dv}{dx} = \frac{1-v}{2}$.

Separating variables: $\frac{2}{1-v} dv = \frac{dx}{x} \implies -2 \ln |1-v| = \ln |x| + C \implies (1-v)^{-2} = C'x \implies (y-x)^2 = Ax$.

19. Solve $x \frac{dy}{dx} - 5y = x^2$:

(a) $y = \frac{x^2}{2} + cx^7$

(b) $y = -\frac{x^2}{3} + cx^5$

(c) $y = \frac{x^5}{2} - cx^7$

(d) $y = \frac{x^2}{2} - cx^5$

Correct Answer: (b) $y = -\frac{x^2}{3} + cx^5$

Explanation: Standard form: $\frac{dy}{dx} - \frac{5}{x}y = x$. Integrating factor: x^{-5} .

$y x^{-5} = \int x \cdot x^{-5} dx = \int x^{-4} dx = -\frac{1}{3} x^{-3} + c \implies y = -\frac{x^2}{3} + cx^5$.

20. Solve $x \frac{dy}{dx} - 3y = 4x$:

(a) $y = -x + cx^2$

(b) $y = -2x + cx^3$

- (c) $y = -\frac{2}{5}x + cx^5$
 (d) $y = -5x + cx^2$

Correct Answer: (b) $y = -2x + cx^3$

Explanation: Rewrite as $\frac{dy}{dx} - \frac{3}{x}y = 4$. Integrating factor is x^{-3} .
 $yx^{-3} = \int 4x^{-3}dx = -2x^{-2} + c \implies y = -2x + cx^3$.

21. Solve $x\frac{dy}{dx} - 4y = x^3$:

- (a) $y = Cx^2 - x^3$
 (b) $y = Cx^4 + x^3$
 (c) $y = Cx^4 - x^3$
 (d) $y = Cx^3 - x^4$

Correct Answer: (c) $y = Cx^4 - x^3$

Explanation: $\frac{dy}{dx} - \frac{4}{x}y = x^2$. Integrating factor $I(x) = x^{-4}$.
 $yx^{-4} = \int x^2x^{-4}dx = \int x^{-2}dx = -x^{-1} + C \implies y = -x^3 + Cx^4$.

22. Solve $(x + 2y)dx + (2x + 3y)dy = 0$:

- (a) $x^2 + 4xy + 3y^2 = c$
 (b) $x^2 - 2xy - 3y^2 = c$
 (c) $x^2 - 4xy + y^2 = c$
 (d) $x^2 - xy - y^2 = c$

Correct Answer: (a) $x^2 + 4xy + 3y^2 = c$

Explanation: This is exact: $M = x + 2y$, $N = 2x + 3y$. $\frac{\partial M}{\partial y} = 2 = \frac{\partial N}{\partial x}$.

Potential function $F(x, y) = \int (x + 2y)dx = \frac{x^2}{2} + 2xy + g(y)$.

$\frac{\partial F}{\partial y} = 2x + g'(y) = 2x + 3y \implies g'(y) = 3y \implies g(y) = \frac{3}{2}y^2$.

Hence $\frac{x^2}{2} + 2xy + \frac{3y^2}{2} = C_1 \implies x^2 + 4xy + 3y^2 = c$.

23. Solve $y' + y = e^x$:

- (a) $y = 2e^{-x} + Ce^x$
 (b) $y = \frac{1}{2}e^{-x} + Ce^x$
 (c) $y = e^x + Ce^{-x}$
 (d) $y = \frac{1}{2}e^x + Ce^{-x}$

Correct Answer: (d) $y = \frac{1}{2}e^x + Ce^{-x}$

Explanation: Integrating factor $I(x) = e^{\int 1dx} = e^x$.

$ye^x = \int e^x \cdot e^x dx = \int e^{2x} dx = \frac{1}{2}e^{2x} + C \implies y = \frac{1}{2}e^x + Ce^{-x}$.

24. Solve $\frac{dy}{dx} + \frac{1}{x}y = xy^2$:

- (a) $y = (cx - x^2)^{-2}$

- (b) $y = (x^2 - cx)^{-2}$
 (c) $y = (cx - x^2)^{-1}$
 (d) $y = \frac{1}{x}(cx - x^2)^{-1}$

Correct Answer: (c) $y = (cx - x^2)^{-1}$

Explanation: This is a Bernoulli equation with $n = 2$. Divide by y^2 : $y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = x$.
 Let $v = y^{-1} \implies \frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$. Substitution yields: $-\frac{dv}{dx} + \frac{1}{x} v = x \implies \frac{dv}{dx} - \frac{1}{x} v = -x$.
 Integrating factor: x^{-1} . Thus $vx^{-1} = \int -1 dx = -x + c \implies v = cx - x^2 \implies y = (cx - x^2)^{-1}$.

25. Solve $\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 3y = 0$, given $y(0) = 9, y'(0) = 5$:

- (a) $y = e^{-3t} + 8e^t$
 (b) $y = \beta_1 e^{-3t} + \beta_2 e^t$
 (c) $y = \beta_1 e^{-3t} + \beta_2 e^{-t}$
 (d) $y = e^{-3t} + 8e^{-t}$

Correct Answer: (a) $y = e^{-3t} + 8e^t$

Explanation: Characteristic equation: $m^2 + 2m - 3 = 0 \implies (m + 3)(m - 1) = 0 \implies m = -3, 1$.

General solution: $y(t) = Ae^{-3t} + Be^t$.

$y(0) = A + B = 9, y'(0) = -3A + B = 5$. Subtracting equations gives $4A = 4 \implies A = 1$, so $B = 8$.

Hence $y = e^{-3t} + 8e^t$.

26. Given $Ay'' + By' + Cy = 0$, if discriminant $B^2 - 4AC > 0$ with distinct roots m_1, m_2 , the general solution is:

- (a) $\beta_1 e^{m_1 x} + \beta_2 e^{m_2 x}$
 (b) $\beta_1 e^{m_1 x} + \beta_2 x e^{m_2 x}$
 (c) $e^{m_1 x}(\beta_1 \sin m_1 x + \beta_2 \cos m_2 x)$
 (d) $\beta_1 + \beta_2(-m_1 x + m_2 x)$

Correct Answer: (a) $\beta_1 e^{m_1 x} + \beta_2 e^{m_2 x}$

Explanation: When the discriminant is strictly positive, the auxiliary equation yields two real and distinct roots $m_1 \neq m_2$. The fundamental solution set is $\{e^{m_1 x}, e^{m_2 x}\}$, yielding $\beta_1 e^{m_1 x} + \beta_2 e^{m_2 x}$.

27. A second order linear ODE $Ay'' + By' + Cy = f(t)$ is homogeneous if and only if:

- (a) $f(t) \neq 0$
 (b) $f(t) > 0$
 (c) $f(t) = 0$
 (d) $f(t) = 0$ or $f(t) > 0$

Correct Answer: (c) $f(t) = 0$

Explanation: By definition, a linear differential equation is homogeneous when the forcing term $f(t)$ on the right-hand side is identically zero.

28. Solve $y'' - 2y' + 5y = 0$, given $y(0) = 5, y'(0) = 5$:

- (a) $y = e^t(5 \cos 2t + 0 \sin 2t) = 5e^t \cos 2t$
- (b) $y = e^t(5 \cos 2t + 2 \sin 2t)$
- (c) $y = e^t(5 \cos 2t - 2 \sin 2t)$
- (d) $y = e^t(2 \cos 2t + 5 \sin 2t)$

Correct Answer: (a) $5e^t \cos 2t$

Explanation: Characteristic equation: $m^2 - 2m + 5 = 0 \implies m = 1 \pm 2i$.

General solution: $y(t) = e^t(A \cos 2t + B \sin 2t)$.

$y(0) = A = 5$.

$y'(t) = e^t(A \cos 2t + B \sin 2t) + e^t(-2A \sin 2t + 2B \cos 2t) \implies y'(0) = A + 2B = 5$.

Since $A = 5, 5 + 2B = 5 \implies B = 0$. Thus $y = 5e^t \cos 2t$.

29. Solve $y'' + 9y' + 20y = 11e^{-4t}$:

- (a) $11te^{-4t}$
- (b) $11te^{-4t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$
- (c) $10e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{-6t}$
- (d) $2e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{-6t}$

Correct Answer: (b) $11te^{-4t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$

Explanation: Homogeneous roots: $m^2 + 9m + 20 = 0 \implies (m + 5)(m + 4) = 0 \implies m = -5, -4$.

So $y_h = \beta_1 e^{-5t} + \beta_2 e^{-4t}$.

Since e^{-4t} is part of y_h , assume particular solution $y_p = Ate^{-4t}$.

$y'_p = Ae^{-4t}(1 - 4t), y''_p = Ae^{-4t}(-8 + 16t)$.

Substitute: $Ae^{-4t}[(-8 + 16t) + 9(1 - 4t) + 20t] = Ae^{-4t}(1) = 11e^{-4t} \implies A = 11$.

Thus $y(t) = 11te^{-4t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$.

30. Particular solution for $y'' + 2y' - 3y = \sin 3t$:

- (a) $\frac{1}{30}(\sin 3t + 2 \cos 3t)$
- (b) $-\frac{1}{30}(2 \sin 3t - \cos 3t)$
- (c) $-\frac{1}{150}(2 \sin 3t + \cos 3t) / -\frac{1}{30}(\dots)$
- (d) $\frac{1}{30}(\sin 3t + 4 \cos 3t)$

Correct Answer: (c) $-\frac{1}{30}(2 \sin 3t + \cos 3t)$ (Standard Form)

Explanation: Let $y_p = A \sin 3t + B \cos 3t$. $y'_p = 3A \cos 3t - 3B \sin 3t$, $y''_p = -9A \sin 3t - 9B \cos 3t$.

Substitute: $(-9A - 6B - 3A) \sin 3t + (-9B + 6A - 3B) \cos 3t = \sin 3t$.

Equating coefficients: $-12A - 6B = 1$ and $6A - 12B = 0 \implies A = 2B$.

$-12(2B) - 6B = 1 \implies -30B = 1 \implies B = -1/30$, $A = -1/15 = -2/30$.

Thus $y_p = -\frac{1}{30}(2 \sin 3t + \cos 3t)$.

31. Particle speed $v = 3t^2 + 4t$ m/s. Find distance traveled at $t = 5$ s given $s(0) = 0$:

- (a) 175 m
- (b) 200 m
- (c) 400 m
- (d) 800 m

Correct Answer: (a) 175 m

Explanation: $s(t) = \int (3t^2 + 4t) dt = t^3 + 2t^2 + C$. Since $s(0) = 0$, $C = 0$.

At $t = 5$: $s(5) = 5^3 + 2(5^2) = 125 + 50 = 175$ m.

32. Particle position $s = 5t^3 + 3t^2$. Find acceleration at $t = 3$ s:

- (a) 14 m/s²
- (b) 96 m/s²
- (c) 100 m/s²
- (d) 90 m/s²

Correct Answer: (b) 96 m/s²

Explanation: Velocity $v(t) = \frac{ds}{dt} = 15t^2 + 6t$.

Acceleration $a(t) = \frac{dv}{dt} = 30t + 6$.

At $t = 3$: $a(3) = 30(3) + 6 = 90 + 6 = 96$ m/s².

33. Velocity vector $\mathbf{v} = (4t + 5, 2t + 3t^2, t + 5t^2)$. Position at $t = 2$ s starting from origin:

- (a) (5, 0, 5)
- (b) (3, 1, 4)
- (c) (18, 12, ...) / closest vector option
- (d) (18, 16, 15.33)

Correct Answer: (18, 12, $\frac{46}{3}$) $\approx$ (18, 12, 15.33)

Explanation: Integrate component-wise with $\mathbf{r}(0) = \mathbf{0}$:

$x(t) = \int (4t + 5) dt = 2t^2 + 5t \implies x(2) = 2(4) + 10 = 18$.

$y(t) = \int (2t + 3t^2) dt = t^2 + t^3 \implies y(2) = 4 + 8 = 12$.

$z(t) = \int (t + 5t^2) dt = \frac{1}{2}t^2 + \frac{5}{3}t^3 \implies z(2) = 2 + \frac{40}{3} = \frac{46}{3} \approx 15.33$.

34. Solve $y'' + 9y' + 20y = 0$:

- (a) $\beta_1 e^{-5t} + \beta_2 e^{-4t}$
- (b) $\beta_1 e^{-4t} + \beta_2 e^{5t}$
- (c) $\beta_1 e^{5t} + \beta_2 e^{6t}$
- (d) $\beta_1 e^{4t} + \beta_2 e^{5t}$

Correct Answer: (a) $\beta_1 e^{-5t} + \beta_2 e^{-4t}$

Explanation: Characteristic equation $m^2 + 9m + 20 = 0 \implies (m + 5)(m + 4) = 0 \implies m = -5, -4$.

Thus, $y = \beta_1 e^{-5t} + \beta_2 e^{-4t}$.

35. Solve $y'' + 4y' + 3y = 15$:

- (a) $\beta_1 e^{-2t} + \beta_2 e^{-4t} + 5x$
- (b) $\beta_1 e^{2t} + \beta_2 e^{-4t} + 15$
- (c) $\beta_1 e^{-3t} + \beta_2 e^{-t} + 5$
- (d) $\beta_1 e^{3t} + \beta_2 e^{-x} + 15x$

Correct Answer: (c) $\beta_1 e^{-3t} + \beta_2 e^{-t} + 5$

Explanation: Characteristic equation $m^2 + 4m + 3 = 0 \implies (m + 3)(m + 1) = 0 \implies m = -3, -1$.

Complementary solution $y_h = \beta_1 e^{-3t} + \beta_2 e^{-t}$.

For particular solution, try constant $y_p = C \implies 3C = 15 \implies C = 5$.

General solution: $y = \beta_1 e^{-3t} + \beta_2 e^{-t} + 5$.